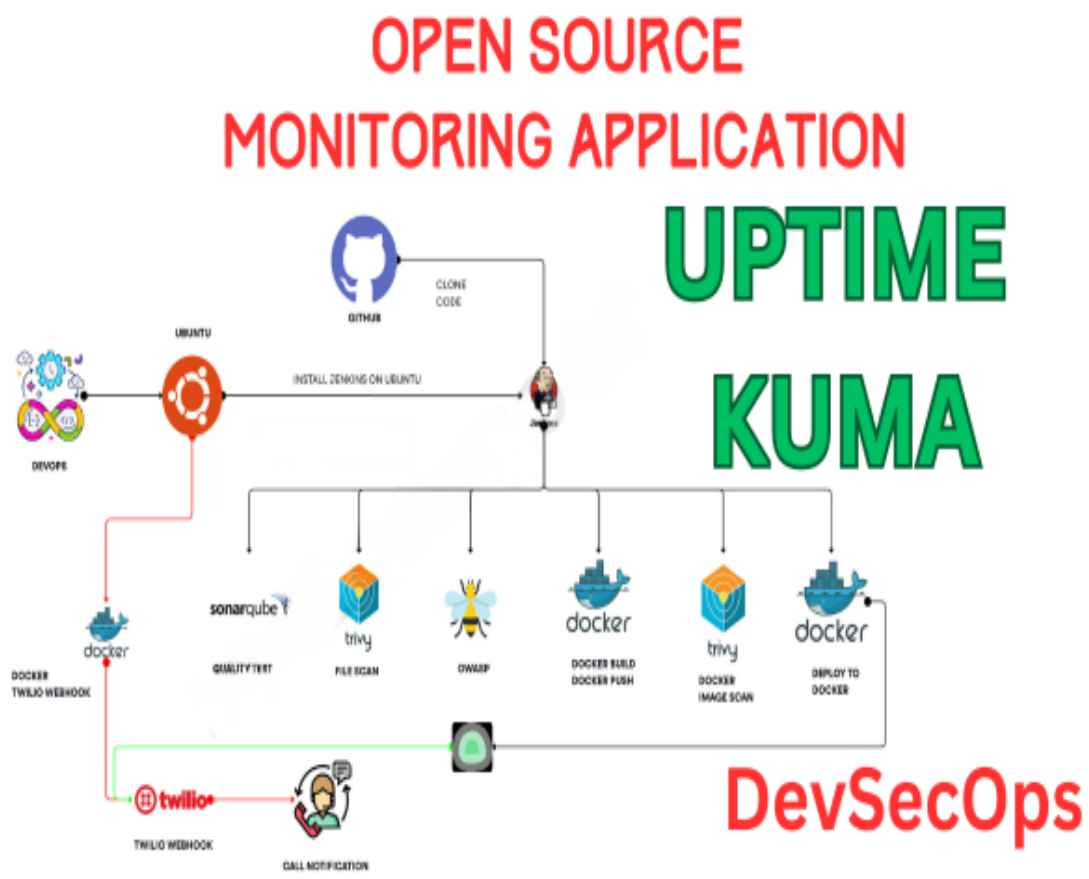


Uptime-Kuma is an open-source self-hosted monitoring tool that provides a sleek interface to track the uptime of websites, services, and APIs. When integrated into a CI/CD pipeline for a blog, it helps ensure continuous availability and performance monitoring. Here's a detailed explanation of how Uptime-Kuma can be utilized effectively in this context:

What is Uptime-Kuma

Overview: Uptime-Kuma is a self-hosted monitoring tool designed to track the uptime and performance of websites, services, and APIs.

Features:

- ✓ Real-time monitoring with customizable intervals.
- ✓ Alerts via multiple channels (email, Slack, Telegram, etc.).
- ✓ Historical data and uptime statistics.
- ✓ Sleek, user-friendly interface.

1. AWS Account and Login

- Log in to the AWS Management Console.

2. Launch an EC2 Instance

A. Navigate to EC2 Dashboard:

- Go to the EC2 Dashboard.

B. Choose an Amazon Machine Image (AMI):

- Click "Launch Instance."
- In the AMI selection screen, search for "Ubuntu 24.04".

C. Choose an Instance Type:

- Select the t2.large instance type.
- Click "Next: Configure Instance Details."

D. Configure Instance Details:

- Use the default settings.
- Ensure you have selected the correct VPC and subnet.
- Click "Next: Add Storage."

E. Add Storage:

- Modify the root volume size to 30 GB.
- Click “Next: Add Tags.”

F. Add Tags:

- (Optional) Add tags for better management.
- Click “Next: Configure Security Group.”

G. Configure Security Group:

- Create a new security group.
- Set the security group name and description.
- Add a rule to allow all traffic (not recommended for production environments):
  - Type: All Traffic
  - Protocol: All
  - Port Range: All
  - **Port Range** : 22, 80, 443, 8080, 9000, 3001, 5000
  - Source: 0.0.0.0/0 (for IPv4) and ::/0 (for IPv6)
- **Warning:** Opening all ports is insecure and should only be done for learning purposes in a controlled environment.
- Click “Review and Launch.”

H. Review Instance Launch:

- Review all configurations.
- Click “Launch.”

Install Jenkins, Docker and Trivy

To Install Jenkins

Connect to your console, and enter these commands to Install Jenkins

```
vi jenkins.sh

#!/bin/bash

sudo apt update -y

wget -O - https://packages.adoptium.net/artifactory/api/gpg/key/public | sudo tee
/etc/apt/keyrings/adoptium.asc

echo "deb [signed-by=/etc/apt/keyrings/adoptium.asc]
https://packages.adoptium.net/artifactory/deb $(awk -F=
'/^VERSION_CODENAME/{print$2}' /etc/os-release) main" | sudo tee
/etc/apt/sources.list.d/adoptium.list

sudo apt update -y

sudo apt install temurin-17-jdk -y

/usr/bin/java --version

curl -fsSL https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key | sudo tee
/usr/share/keyrings/jenkins-keyring.asc > /dev/null

echo deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc]
https://pkg.jenkins.io/debian-stable binary/ | sudo tee /etc/apt/sources.list.d/jenkins.list >
/dev/null

sudo apt-get update -y

sudo apt-get install jenkins -y

sudo systemctl start jenkins

sudo systemctl status jenkin

sudo chmod 777 jenkins.sh

sudo su   #move into root and run

./jenkins.sh    # this will installl jenkins
```

Once Jenkins is installed, you will need to go to your AWS EC2 Security Group and open Inbound Port 8080, since Jenkins works on Port 8080.

Now, grab your Public IP Address

<EC2 Public IP Address:8080>

sudo cat /var/lib/jenkins/secrets/initialAdminPassword

Unlock Jenkins using an administrative password and install the suggested plugins.

Jenkins will now get installed and install all the libraries.

Create a user click on save and continue.

Jenkins Getting Started Screen.

Install Docker

Add Docker's official GPG key:

sudo apt-get update

sudo apt-get install ca-certificates curl

sudo install -m 0755 -d /etc/apt/keyrings

sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc

sudo chmod a+r /etc/apt/keyrings/docker.asc

Add the repository to Apt sources:

```
echo \
  "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.asc] https://download.docker.com/linux/ubuntu \
  $(. /etc/os-release && echo "$VERSION_CODENAME") stable" | \
  sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
sudo apt-get update
sudo apt-get install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin -y
sudo usermod -aG docker ubuntu
newgrp docker
sudo chmod 777 /var/run/docker.sock
```

After the docker installation, we create a sonarqube container (Remember to add 9000 ports in the security group).

```
sudo chmod 777 /var/run/docker.sock
docker run -d --name sonar -p 9000:9000 sonarqube:lts-community
```

Install Trivy

```
vi script.sh
sudo apt-get install wget apt-transport-https gnupg lsb-release -y
wget -qO - https://aquasecurity.github.io/trivy-repo/deb/public.key | gpg --dearmor | sudo tee /usr/share/keyrings/trivy.gpg > /dev/null
echo "deb [signed-by=/usr/share/keyrings/trivy.gpg] https://aquasecurity.github.io/trivy-repo/deb $(lsb_release -sc) main" | sudo tee -a /etc/apt/sources.list.d/trivy.list
```

sudo apt-get update

sudo apt-get install trivy –y

Give permissions and run script

sudo chmod 777 script.sh

./script.sh

Next, we will log in to Jenkins and start to configure our Pipeline in Jenkins

Install Plugins like JDK, Sonarqube Scanner, NodeJs, OWASP Dependency Check

Install Plugin

Goto Manage Jenkins →Plugins → Available Plugins →

Install below plugins

Blue ocean

1 → Eclipse Temurin Installer

2 → SonarQube Scanner

3 → NodeJs Plugin

4 → Docker

5 → Docker commons

6 → Docker pipeline

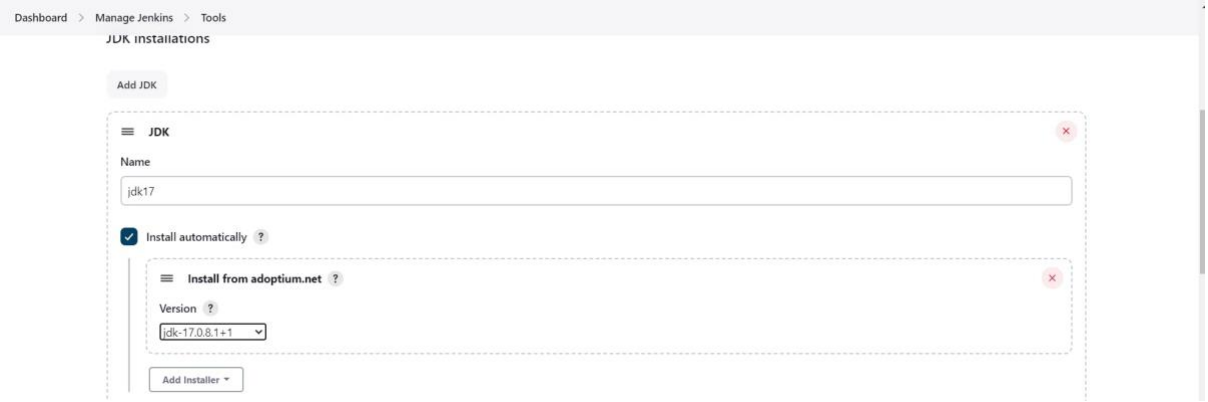
7 → Docker API

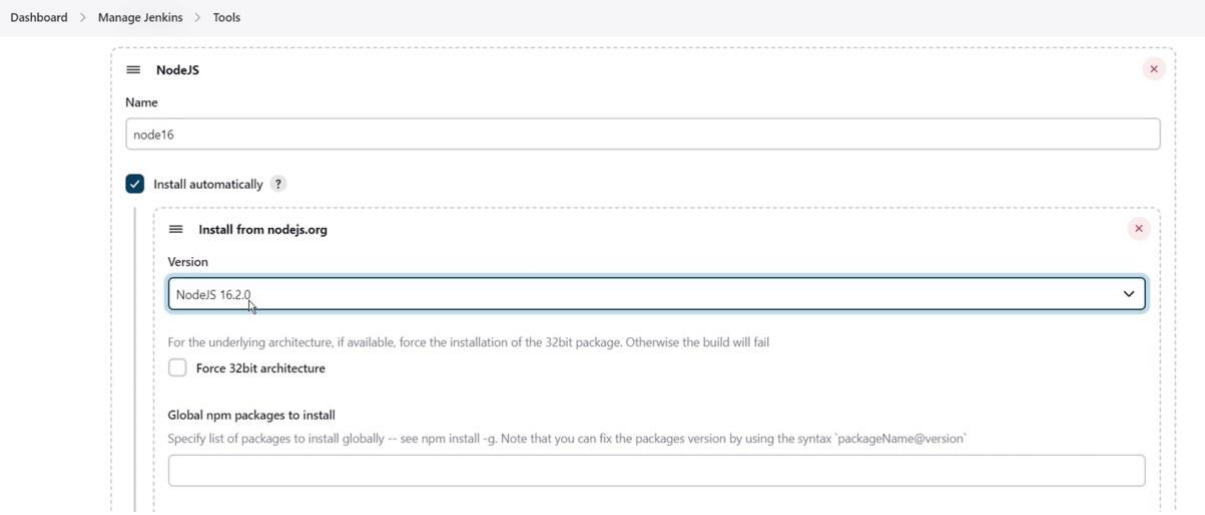
8 → Docker Build step

9 → Owasp Dependency Check

Configure Java and Nodejs in Global Tool Configuration

Goto Manage Jenkins → Tools → Install JDK(17) and NodeJs(18)→ Click on Apply and Save



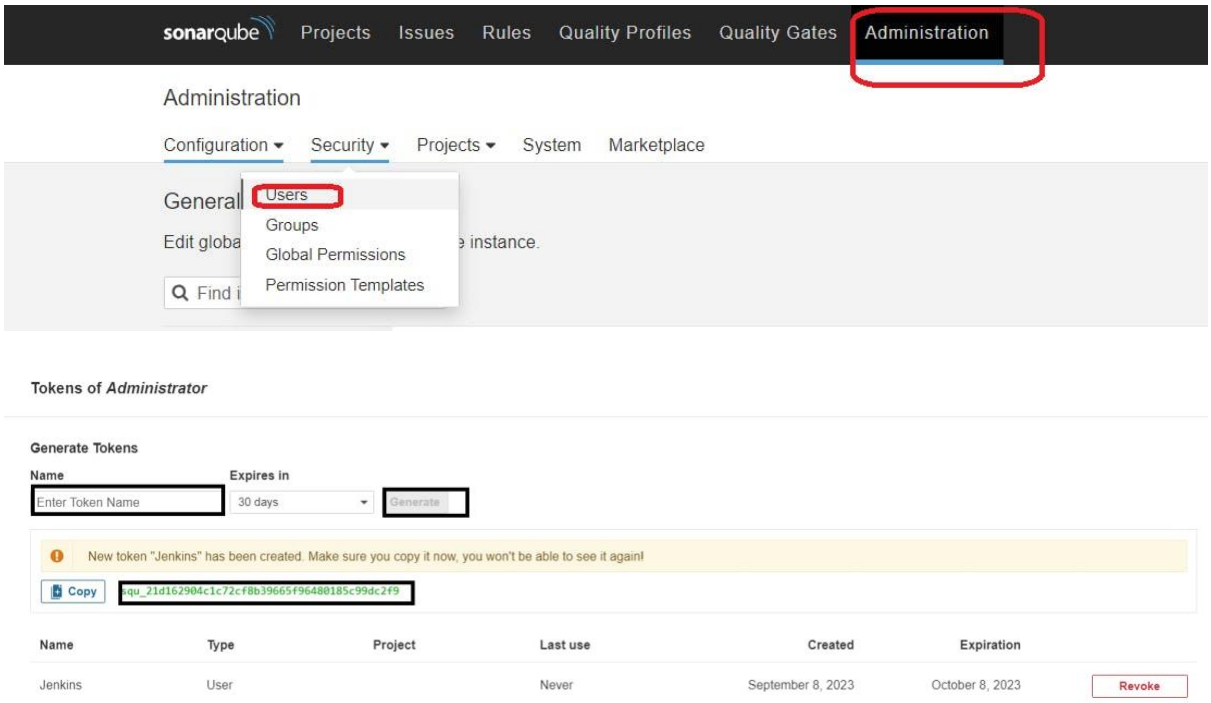


Grab the Public IP Address of your EC2 Instance, Sonarqube works on Port 9000, so <Public IP>:9000. Goto your Sonarqube Server.

Click on Administration → Security → Users → Click on Tokens and Update Token → Give it a name → and click on Generate Token

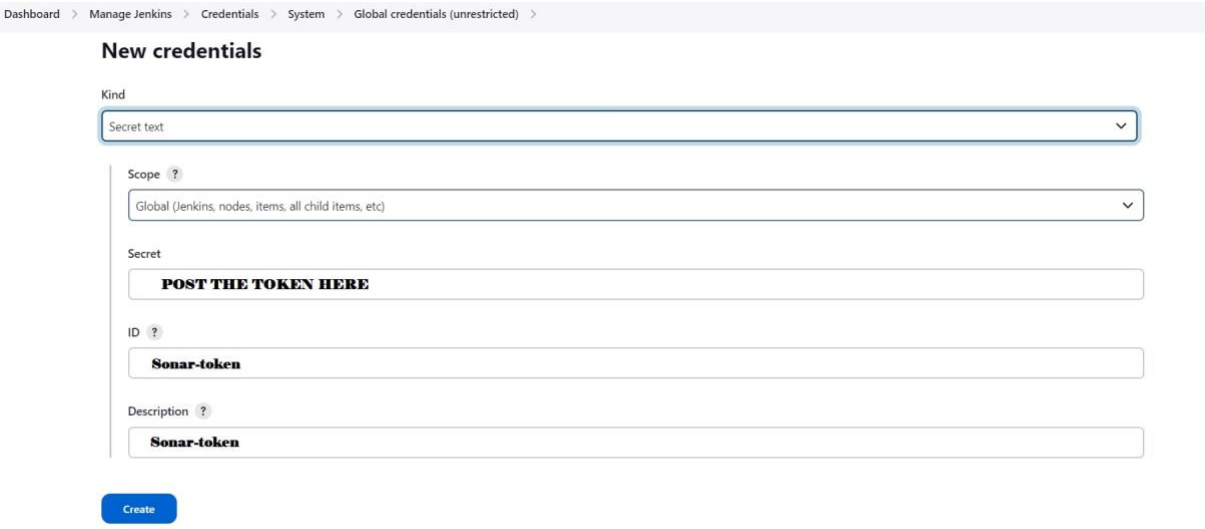
Step 6 — Configure Sonar Server in Manage Jenkins

Grab the Public IP Address of your EC2 Instance, Sonarqube works on Port 9000, so <Public IP>:9000. Goto your Sonarqube Server. Click on Administration → Security → Users → Click on Tokens and Update Token → Give it a name → and click on Generate Token



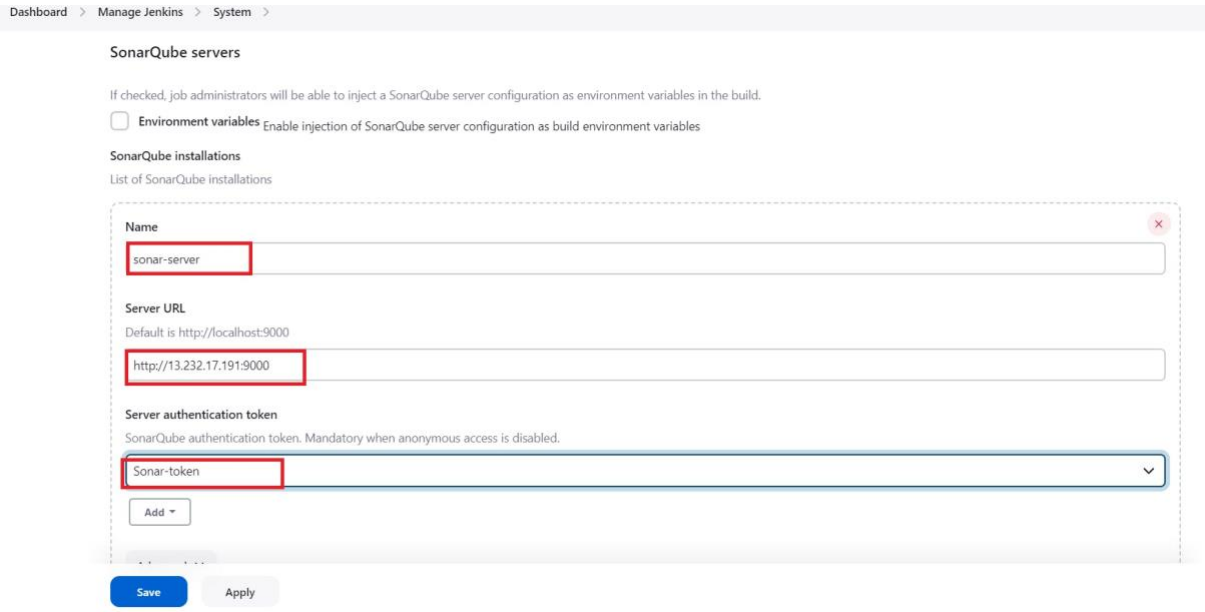
copy Token

Goto Jenkins Dashboard → Manage Jenkins → Credentials → Add Secret Text. It should look like this



You will this page once you click on create

Now, go to Dashboard → Manage Jenkins → System and Add like the below image.

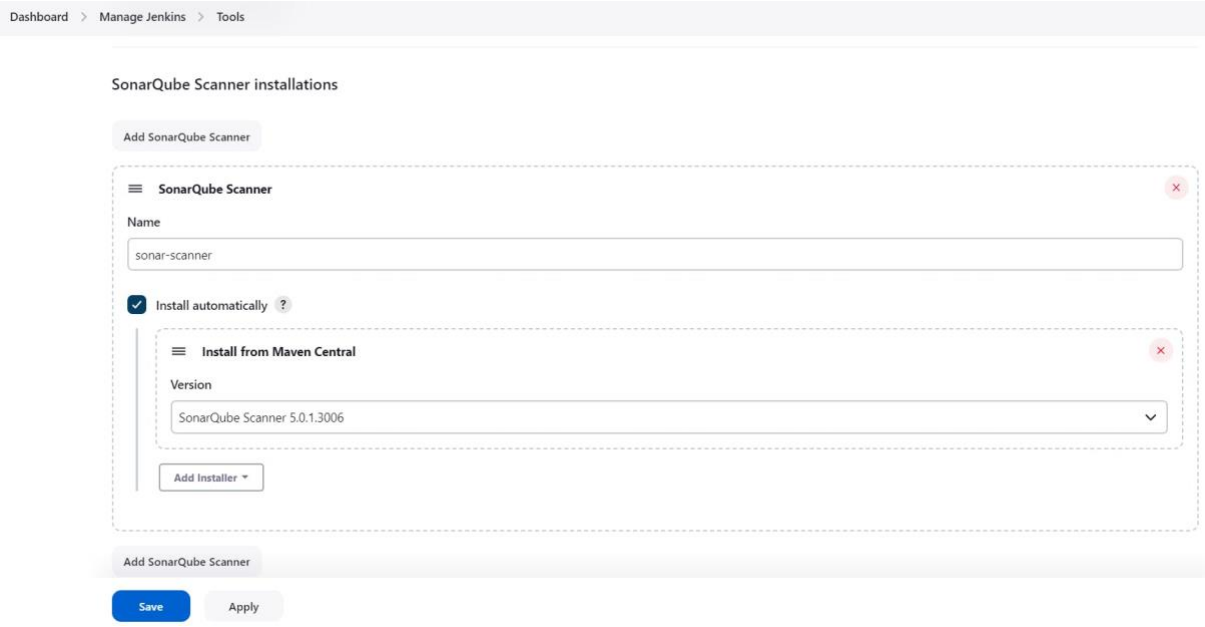


Click on Apply and Save

The Configure System option is used in Jenkins to configure different server

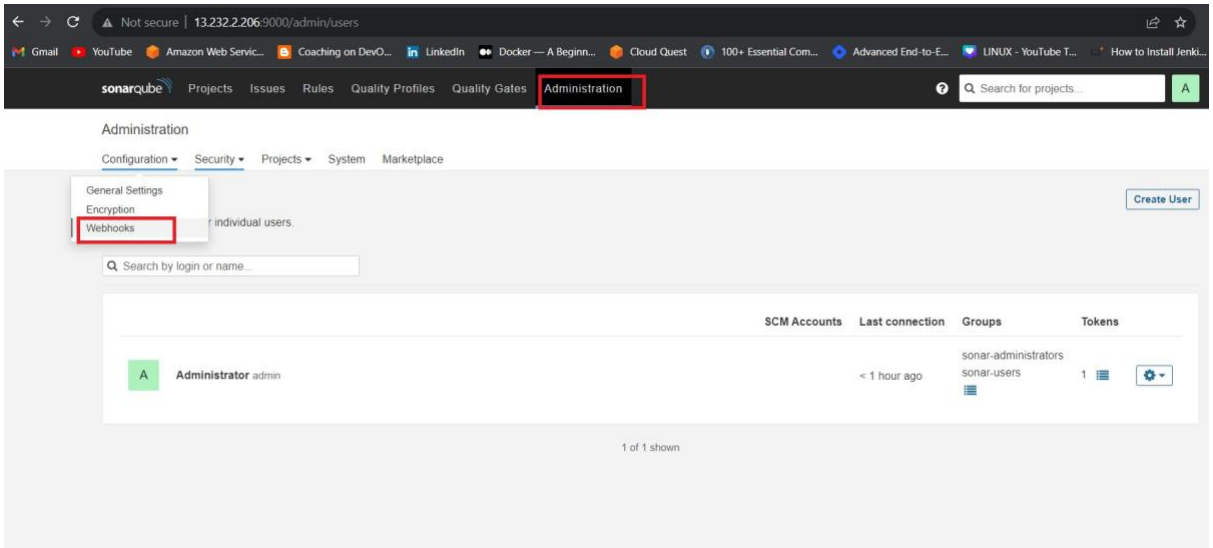
Global Tool Configuration is used to configure different tools that we install using Plugins

We will install a sonar scanner in the tools.

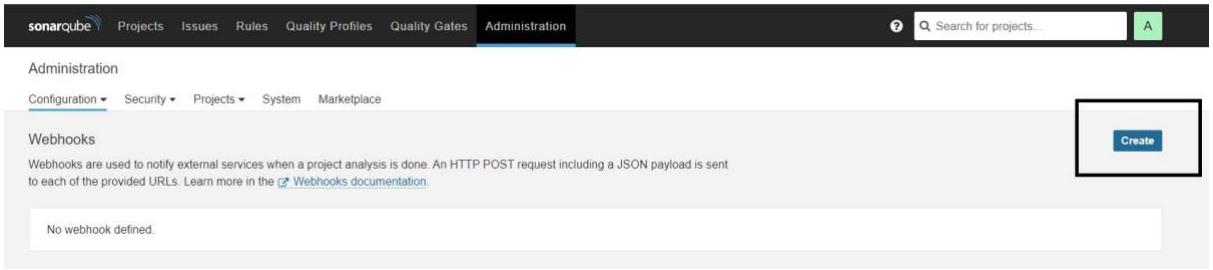


In the Sonarqube Dashboard add a quality gate also

Administration–> Configuration–>Webhooks



Click on Create



Add details

#in url section of quality gate

<http://jenkins-public-ip:8080>/sonarqube-webhook/

First, we configured the Plugin and next, we had to configure the Tool

Goto Dashboard → Manage Jenkins → Tools →Dependency-Check & Docker

Go to manage Jenkins –> Credentials

Add DockerHub Username and Password under Global Credentials

Create Job for Uptime-kuma

Add this stage to Pipeline Script

```
pipeline{
   agent any
   tools{
      jdk 'jdk17'
      nodejs 'node18'
   }
   environment {
      SCANNER_HOME=tool 'sonar-scanner'
```

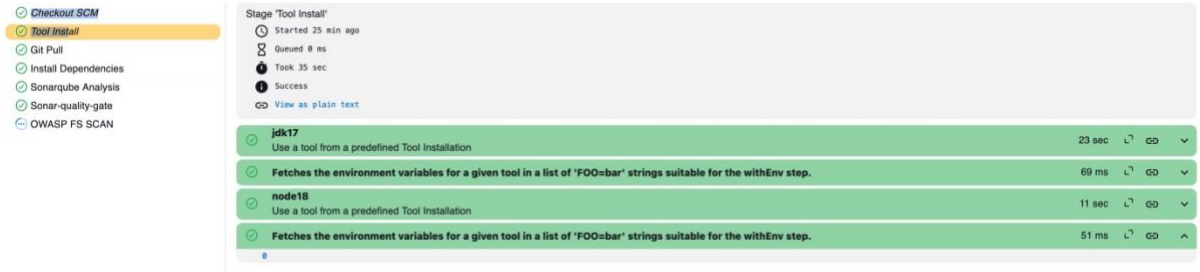
```
}
stages {
    stage('Checkout from Git'){
        steps{
            git branch: 'main', url: 'https://github.com/Milky19/uptime.git'
        }
    }
    stage('Install Dependencies') {
        steps {
            sh "npm install"
        }
    }
    stage("Sonarqube Analysis "){
        steps{
            withSonarQubeEnv('sonar-server') {
                sh ''' $SCANNER_HOME/bin/sonar-scanner -Dsonar.projectName=uptime \
                -Dsonar.projectKey=uptime '''
            }
        }
    }
    stage("quality gate"){
       steps {
```

```
            script {
                waitForQualityGate abortPipeline: false, credentialsId: 'Sonar-token'
            }
        }
    }
    stage('OWASP FS SCAN') {
        steps {
            dependencyCheck additionalArguments: '--scan ./ --disableYarnAudit --
disableNodeAudit', odcInstallation: 'DP-Check'
            dependencyCheckPublisher pattern: '**/dependency-check-report.xml'
        }
    }
    stage('TRIVY FS SCAN') {
        steps {
            sh "trivy fs . > trivyfs.json"
        }
    }
    stage("Docker Build & Push"){
        steps{
            script{
                withDockerRegistry(credentialsId: 'docker', toolName: 'docker'){
                    sh "docker build -t uptime ."
                    sh "docker tag uptime hanvitha/uptime:latest "
```

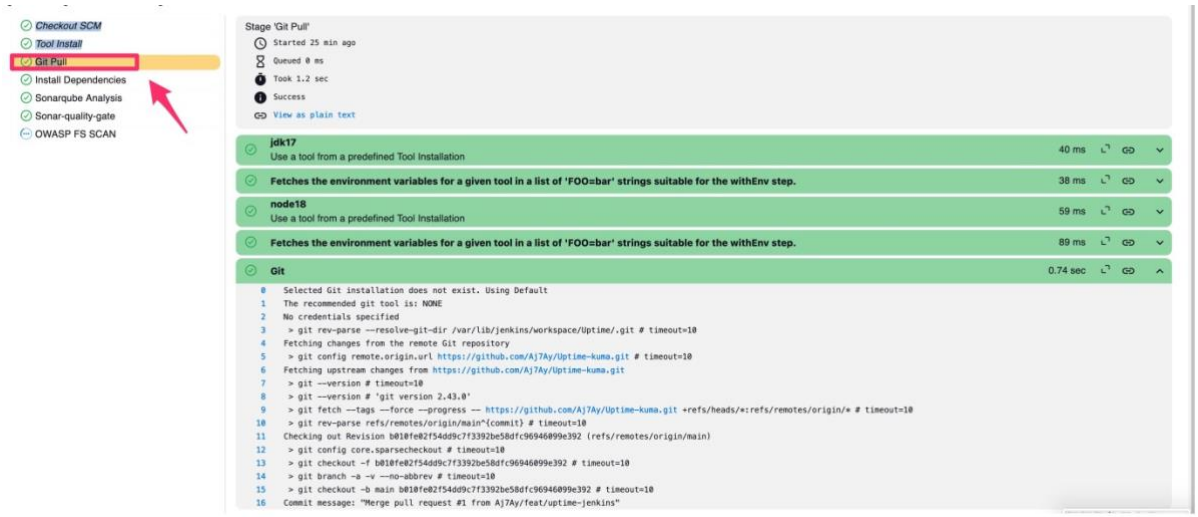
```
                sh "docker push hanvitha/uptime:latest "
            }
        }
    }
}
stage("TRIVY"){
    steps{
        sh "trivy image sevenajay/uptime:latest > trivy.json"
    }
}
stage ("Remove container") {
    steps{
        sh "docker stop uptime | true"
        sh "docker rm uptime | true"
     }
}
stage('Deploy to container'){
    steps{
        sh 'docker run -d --name chatbot -v /var/run/docker.sock:/var/run/docker.sock -p
3001:3001 hanvitha/uptime:latest'
    }
}
}
```

OUTPUT OF PIPELINE

EACH STEP IMAGES

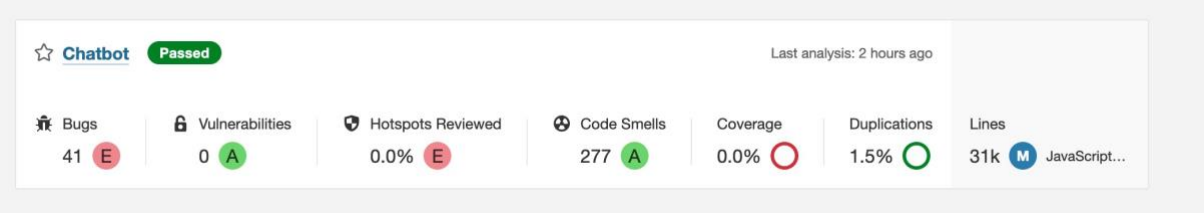


GIT INSTALL

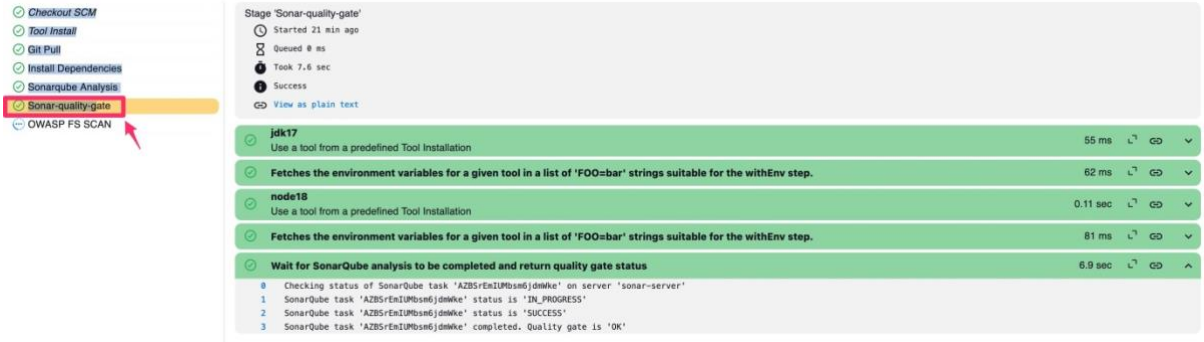


NPM install

Sonarqube analysis

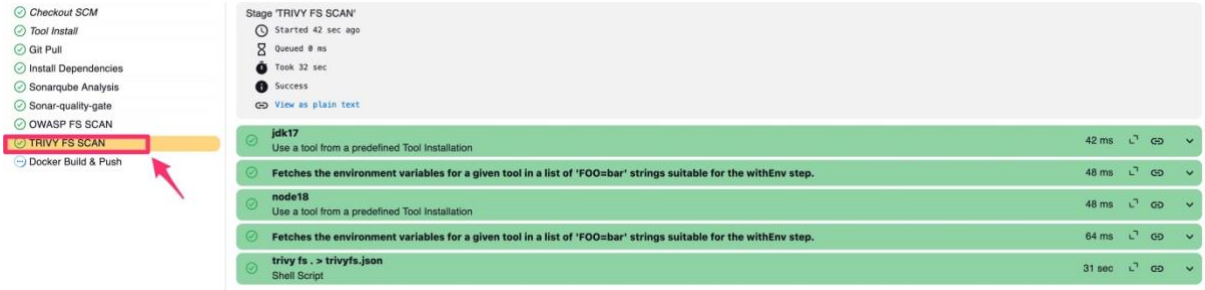


Quality Gate



Owasp

Trivy FS

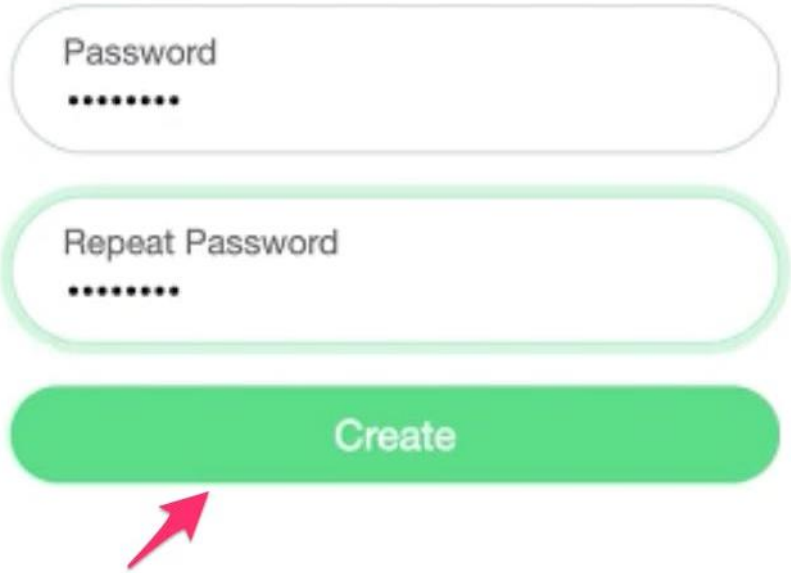


Trivy Image scan and deploy to Docker

Copy the Public of Server

<public-ip>:30001

Output Login Page



Twilio webhook setup

Clone the Github Repo

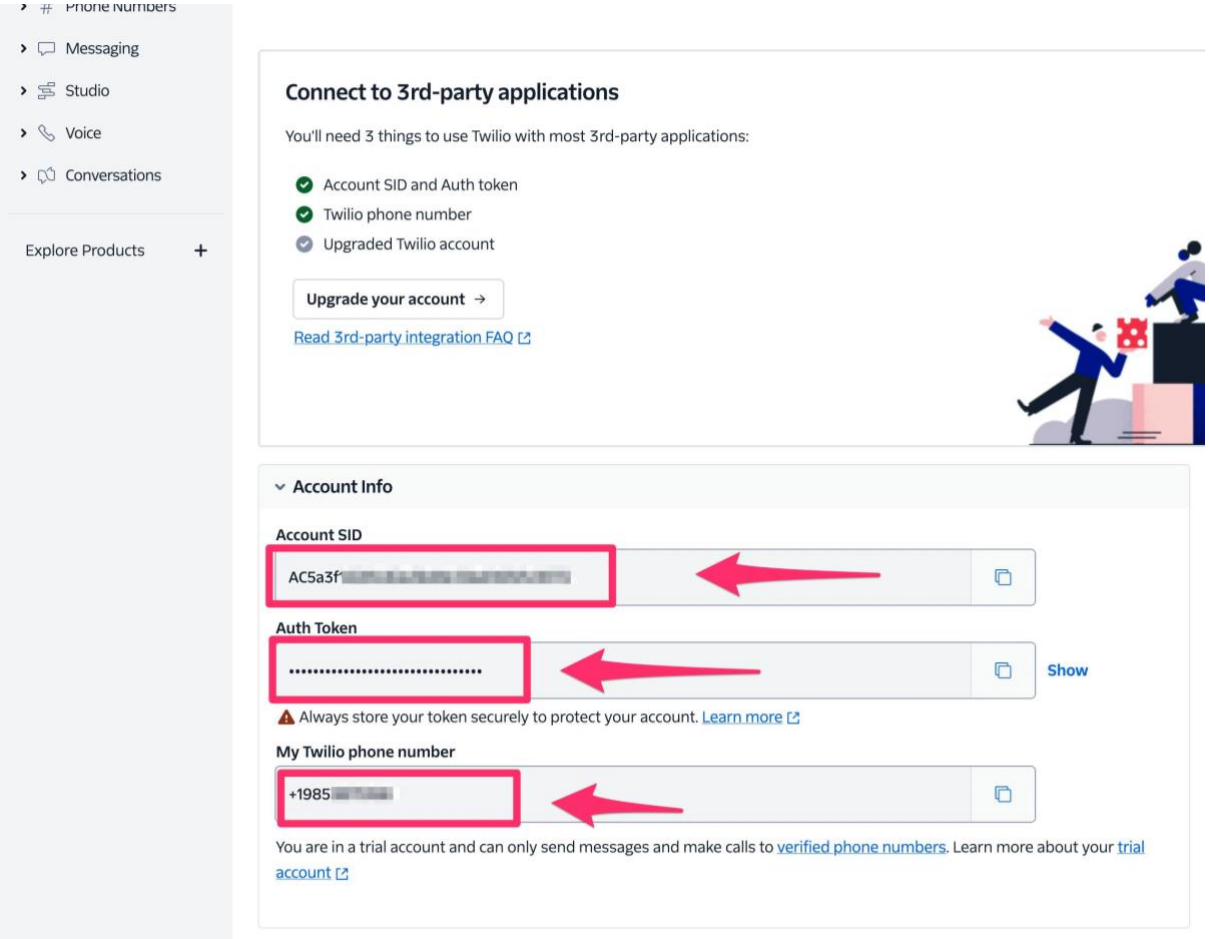
git clone https://github.com/Milky19/Uptime-kuma.git

cd Uptime-kuma

cd Webhook

Go to Console

copy Account SID & Account Auth Token and Mobile Number



Update the [app.py]

```
from flask import Flask

from twilio.rest import Client

app = Flask(__name__)

@app.route('/trigger-calls', methods=['GET', 'POST'])

def trigger_call():

    account_sid = '<acc-sid>'

    auth_token = '<acc-auth>'
```

```
    client = Client(account_sid, auth_token)

    # List of numbers to call

    numbers_to_call = ['<country-code-number>', '<country-code-number>']

    call_sid = []  # To store Call SIDs

    for number in numbers_to_call:

        call = client.calls.create(

                    twim='<Response><Say>"change Alert message here"</Say></Response>',

                    to=number,

                    from_='<twilio-number>'

                )

        call_sids.append(call.sid)

    # Joining all the Call SIDs to return in response

    call_sids_str = ', 'join(call_sids)

    return f"Call initiated with SIDs: {call_sids_str}", 200

if __name__ == '__main__':

    app.run(host'0.0.0.0', port=5000)
```

Now install These on Server

```
sudo apt install python3-pip

sudo apt install python3.12-venv

python3 -m venv venv

source venv/bin/activate
```

Now install Requirements

pip install -r requirements.txt

Create the Docker webhook

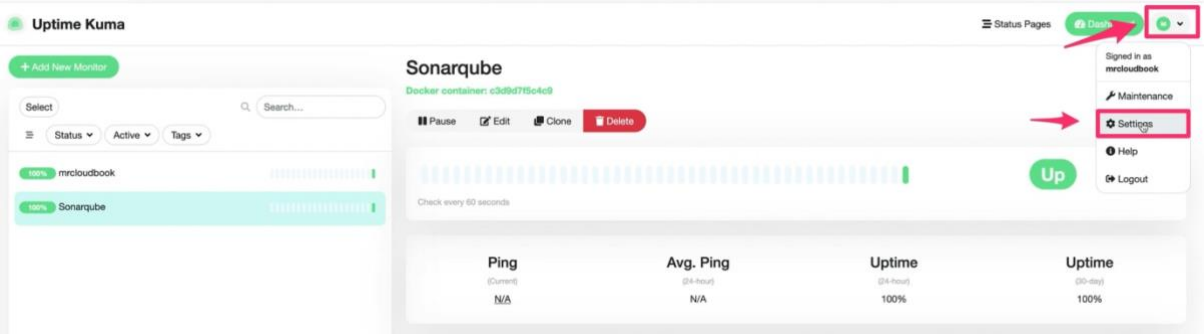
cd Uptime-kuma/webhook

docker build -t webhook .

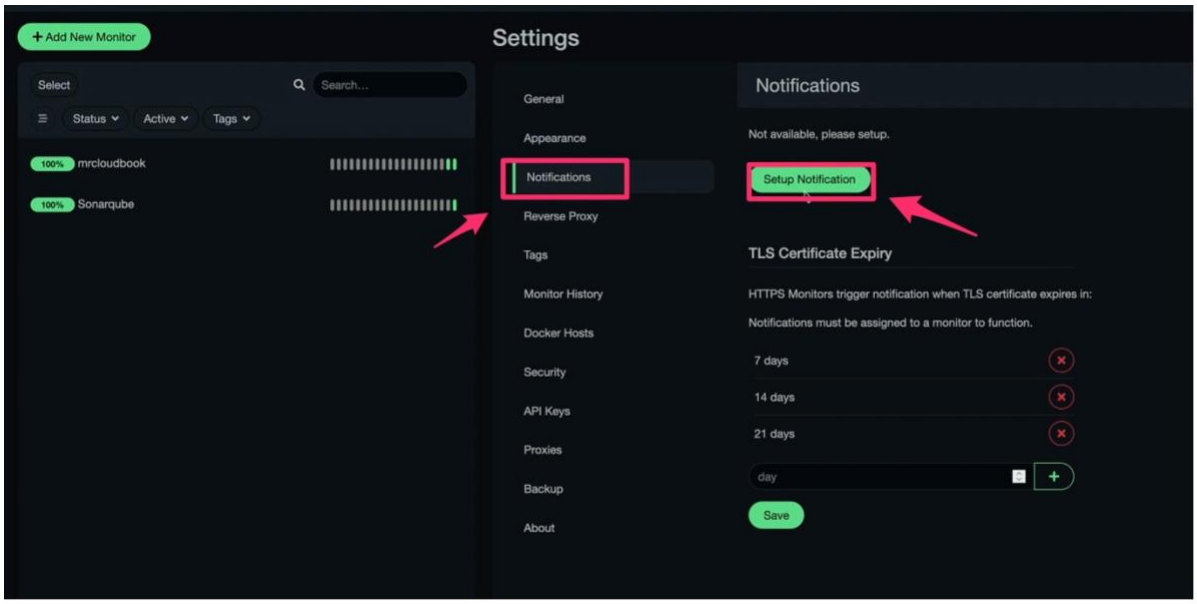
docker run -d --name webhook -p 5000:5000 webhook:latest

webhook runs on 5000 port

Go to Uptime Kuma select Profile –> Settings

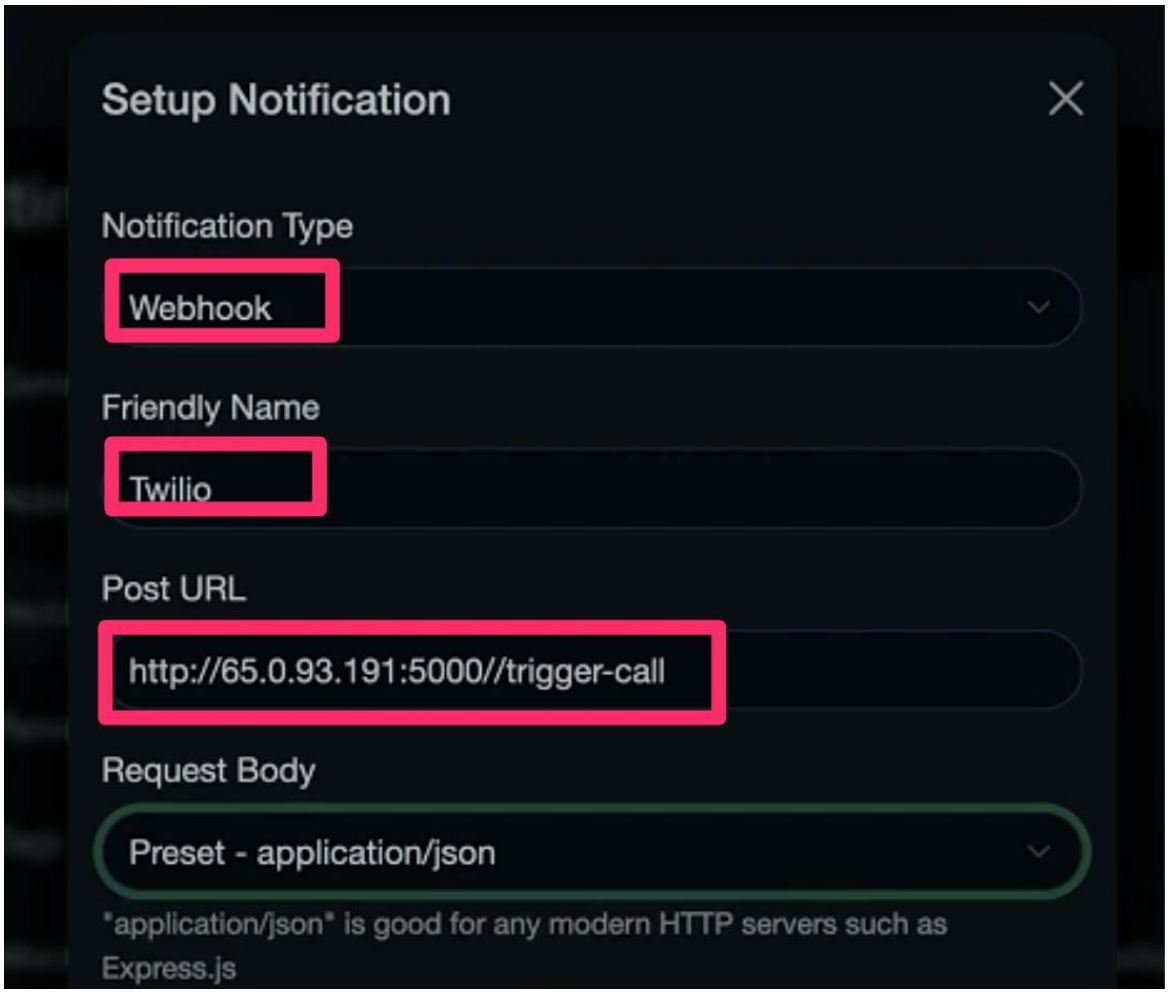


Select Notification –> setup Notification

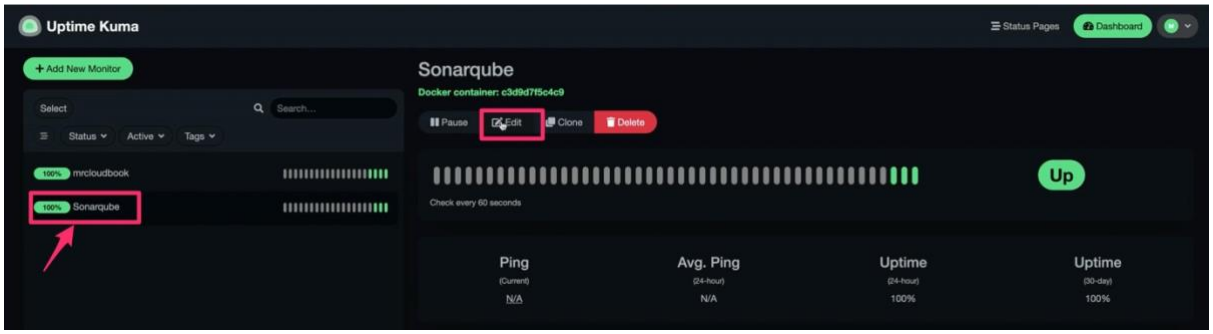


Select Notification Type as webhook

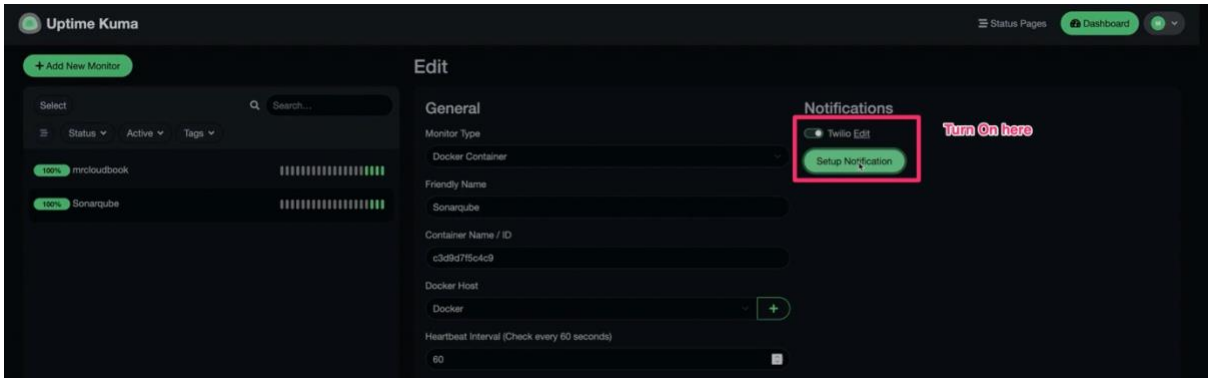
Give Friendly name –> URL of webhook –> save



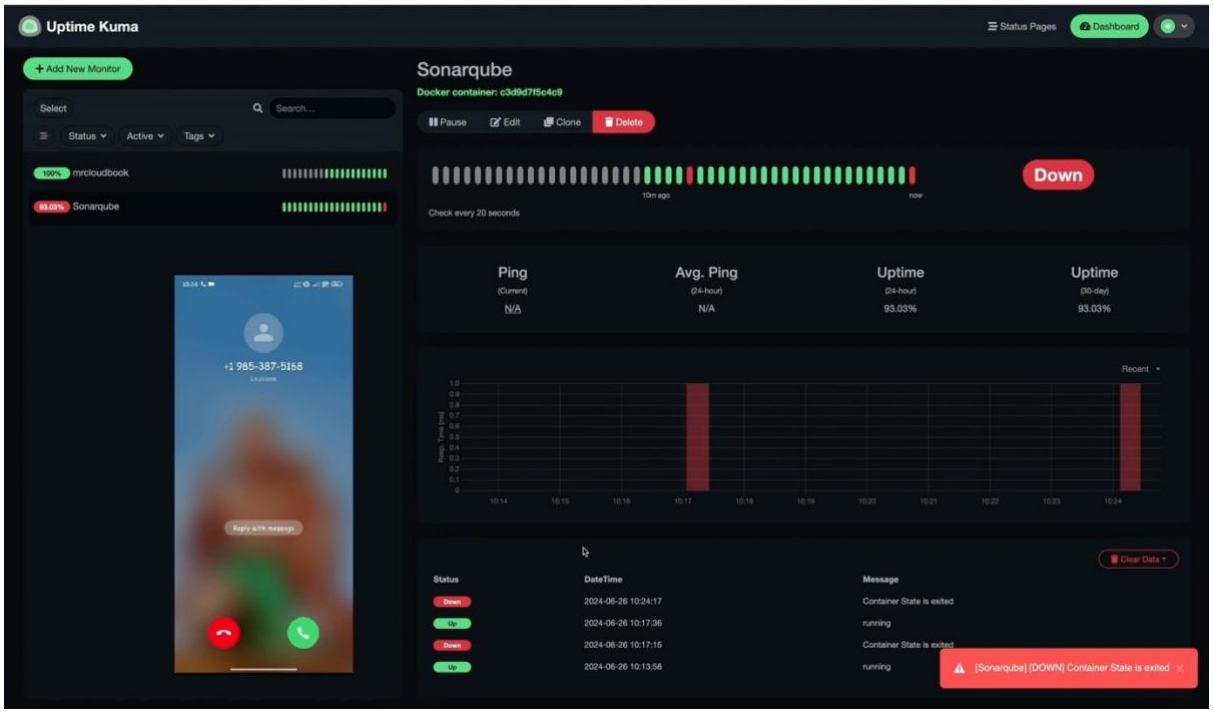
Select One service from Dashboard –> Edit



Turn on Notification to get call



You will get call when server or service Down



Remove Server at end.